2024

BILL Uses Amazon ECS with AWS Fargate to Effectively Scale Applications

Learn how financial operations compan
Amazon ECS with AWS Fargate.

Overview | Opportunity | Solution | Outcome | AWS Services Used | Architecture Diagram

Significant reduction

in operational cost savings

150,000–200,000 requests

met per minute with automatic scaling

Ability to spin up

new environments within weeks instead of months

Increased

developer productivity with improved tools

Overview

To better accommodate growth and scale over time, financial operations company BILL wanted to migrate the on-premises architecture of its platform to Amazon Web Services (AWS) to refactor processes and increase speed and efficiency. The company began exploring services like Amazon Elastic Container Service (Amazon ECS), a fully managed container orchestration service that simplifies the deployment, management, and scaling of containerized applications.

Using services like Amazon ECS with AWS Fargate, a serverless, pay-as-you-go compute engine that organizations use to focus on building applications without managing servers, BILL improved elasticity and shifted the burden of managing servers to AWS so that its development teams can focus on growth and innovation.

Opportunity | Using Amazon ECS with AWS Fargate to Refactor and Simplify the Architecture of BILL's Core Applications

As a leading financial operations platform for small and midsize businesses (SMBs), BILL's integrated platform helps businesses more efficiently control their payables, receivables, and spend and expense management. Hundreds of thousands of businesses rely on BILL's proprietary member network of millions of businesses to pay or get paid faster. BILL is also a trusted partner of leading US financial institutions, accounting firms, and accounting software providers.

BILL previously ran static web server farms continuously. However, the company wanted to migrate workloads to the cloud to better manage its volume and provide more flexibility. As BILL is at the center of customers' day-to-day financial operations, the company strives to continuously innovate their technology to better serve their customers, while ensuring seamless updates to its daily operations. For the initial cloud project, BILL migrated its cell-based architecture to Amazon Elastic Compute Cloud (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload.

Since then, BILL has continued investing in cloud technology to refactor its architecture. In 2021, the company started migrating its core applications to use Amazon ECS with AWS Fargate. BILL chose a calculated rollout for the migration, dedicating 1 quarter to develop and 1.5 quarters to roll out the changes incrementally. "Now that we've migrated our core applications to Amazon ECS with AWS Fargate, we have a lot of flexibility and can build other microservices using the same blueprints," says Arvind Sharma, Senior Engineering Manager at BILL. "Our developers can move with speed and efficiency to deliver innovation for customers and microservices quickly."

Throughout the process, BILL received support from AWS account and specialist teams to implement the migration successfully and maintain the high security standards required for a financial operations company. "The support we've received from the AWS teams has been awesome and always available when we've needed it," says Nick Thompson, Senior Staff Software Engineer at BILL. "It has saved us time, which is ultimately better efficiency for the company in terms of resources."

Solution | Increasing Developer Productivity While S e

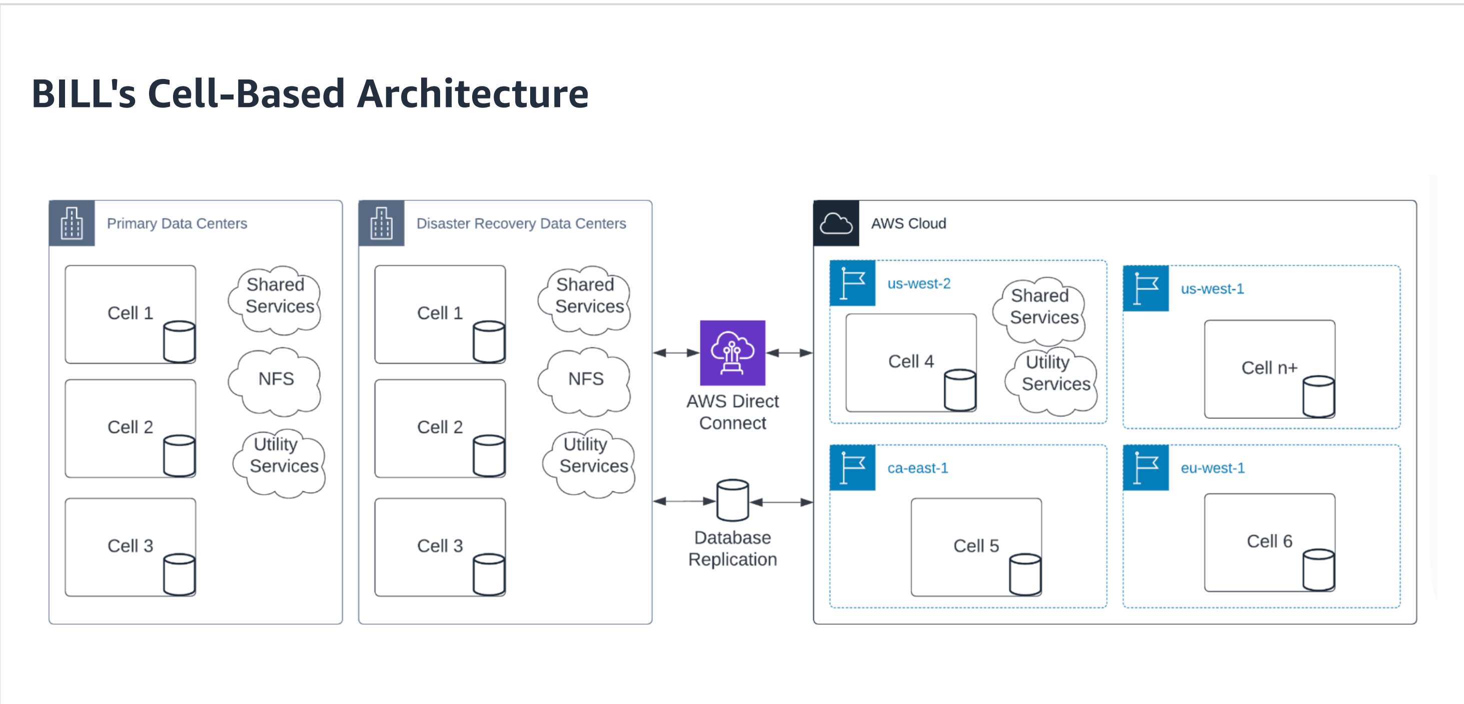
BILL’s front-end application can sometimes receive around 150,000–200,000 requests per minute, so flexible scalability is essential to reduce latency and maintain transaction accuracy for customers. In addition, the company also needs to manage traffic spikes, specifically during the day because most billing transactions occur during business hours. “Using Amazon ECS with AWS Fargate, we can scale up in no time without manual intervention,” says Sharma.

With a serverless solution, BILL also manages future growth more efficiently. BILL can bring larger customers or partners onboard quickly and simply, scaling in a cost-effective way that wouldn’t have been possible with an on-premises infrastructure.

As the company looks to serve more SMBs globally, BILL will need to comply with different regulations for some countries that will require a regional presence to do business. Using Amazon ECS with AWS Fargate, BILL can quickly spin up cloud resources in other AWS Regions using blueprints. “If we need to bring up a new environment in a different country as we support customers internationally, it’s much simpler,” says Sharma. “We’ve achieved significant time savings in our operations. It used to take a couple of months and now takes a few weeks.”

By using a fully managed serverless solution, BILL saves on operational costs and reduces complexity for its technical staff so that they can focus on development using on-demand testing environments. BILL is also gaining time savings in operations by using AWS Fargate Spot, which facilitates running interruption-tolerant Amazon ECS tasks at up to a 70 percent discount compared with AWS Fargate prices, for its nonproduction environments. “Because AWS Fargate is serverless, we can expand internationally as business picks up and everything scales up automatically,” says Sharma. “We also don’t have to worry about the operational cost. We build it, and the serverless AWS solution takes care of everything.”

Architecture Diagram



Outcome | Investing More in a Serverless AWS Solution

Now that the migration of the company’s core applications is complete, BILL is focusing on m al applications to the cloud. BILL plans to continue investing in its serverless solution to reduce c ead and increase efficiency. “We’ve improved velocity, achieved efficiencies, and refactored our architecture by migrating from an on-premises architecture to Amazon ECS with AWS Fargate,” says Subbu Allamaraju, Vice President of Engineering at BILL.

About BILL

BILL is a leading financial operations platform for small and midsize businesses. BILL’s integrated platform helps businesses more efficiently control their payables, receivables, and spend and expense management.

AWS Services Used

Amazon ECS

Amazon Elastic Container Service (Amazon ECS) is a fully managed container orchestration service that simplifies your deployment, management, and scaling of containerized applications.

Learn more »

AWS Fargate

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers. AWS Fargate is compatible with both Amazon Elastic Container Service (Amazon ECS) and Amazon Elastic Kubernetes Service (Amazon EKS).

Learn more »

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud. It is designed to make web-scale cloud computing easier for developers.

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